1. Strong-rho canonicalization, round 2 (2026-10-16)

Study: `research/rho_canon2_20261016/` · **Protocol:** `PROTOCOL.md` (declared before the changes; two addenda, one recorded deviation) · **Baseline:** `72843c83a` (`main`) · **Host:** one x86-64 cloud VM (Intel Xeon @ 2.10 GHz, 4 vCPU), no hardware PMU.

The strong rho is the reference every Koblitz `vs_rho` comparison prices index calculus against. This round makes its canonicalization cheaper without changing any walk. Every output line at n = 41, 53 and 61 is identical to the baseline's, timing aside, and all 13 strong-rho unit tests pass with debug assertions on.

1.1. Changes

1. `MulMod::mul` casts through `i64`: one instruction per conversion instead of a multi-instruction `u64↔f64` sequence.
2. `rotate` drops its `k = 0` branch. A `debug_assert` pins $v < 2^n$, $k \le n$.
3. `canonicalize` reuses the rotations the least-rotation search and the sign test already computed.
4. The orbit multipliers $\pm\lambda^k$ are precomputed with their signs.
5. Shoup's fixed-operand product for $a \cdot w \bmod m$: exact for $a, w < m < 2^{63}$.
6. `canonicalize` is inlined into `solve`.

1.2. Instructions (measured, Callgrind, n = 53, 4 fixtures)

arm	instructions	per walk step
baseline	2,847,169,163	938
changes 1–4	2,580,125,642	850
changes 1–5	2,557,912,731	843
final, 1–6	**2,369,298,800**	**781**

The final build runs 16.8% fewer instructions per step (1.20×), with an identical output digest in every arm.

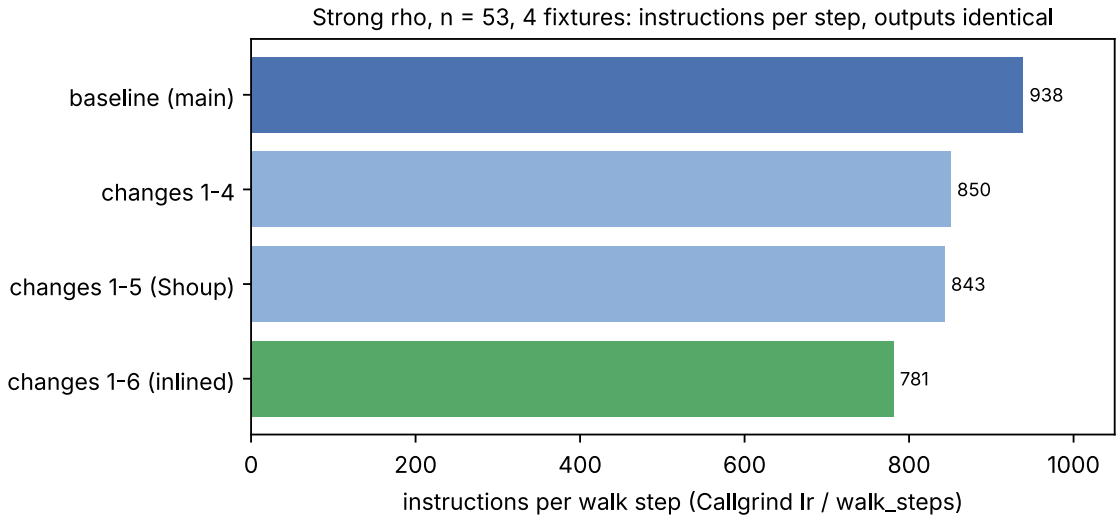


Figure 1: Instructions per walk step by arm (`runs/callgrind/`).

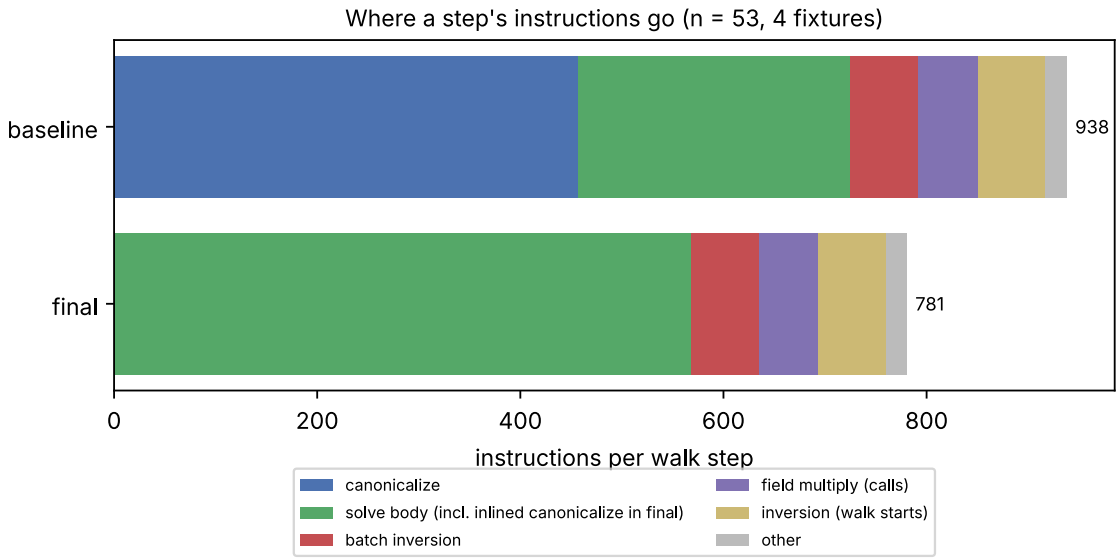


Figure 2: Where a step's instructions go, baseline vs final.

1.3. Native time (measured)

A/A (baseline × 6), then 5 interleaved A/B rounds. 32 processes under `isolated_bench`, none contended.

curve	baseline ns/step	final ns/step	ratio	A/A spread, verdict
`icv1-f2m53-…`	130.8	111.5	1.173	20.6%: not established
`icv1-f2m61-…`	139.7	123.1	1.135	12.1%: outside the spread

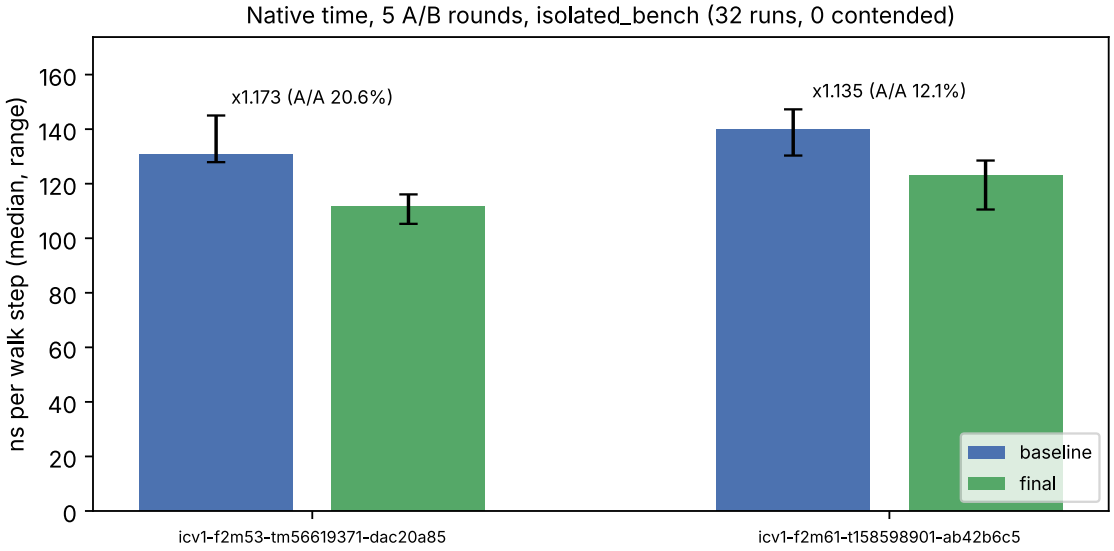


Figure 3: Native ns per walk step, median and range (`runs/time/`).

IPC: the host has no hardware PMU. `perf_ipc.sh` recorded `unsupported`, so no IPC is claimed.

1.4. Scope

- **What remains.** The step is about 37 field multiplications' worth against a floor near 6–7. Most of it is four basis conversions, the rotation search, the `solve` body and the walk-start inversions. Going further would change the walk definition (pinned by committed replays) or need cache-cost measurement this host cannot make.
- **Counted units.** GAE records do not change. Future wall-time and instruction comparisons against this rho get harder for index calculus.